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Pathways to decarbonise the power sector in Mauritius

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ABSTRACT

An electricity modelling study is presented to support the decarbonisation of the power sector in Mauritius. The gaps in literature that this study aims to address are the insufficient evidence of systematic modelling in the country's electricity transition strategies, outdated targets in existing modelling studies, limited long-term climate ambitions and the under-representation of the impact of vehicle electrification on electricity planning. Hence, a techno-economic optimisation model is developed to explore three long-term (2025–2050) scenarios in the power sector of Mauritius. Scenario A reflects the current policy ambition, aligning with the Nationally Determined Contribution (NDC) targets of achieving 60 % of renewable energy in the electricity by 2030. Scenario B models a pathway to achieve net-zero emissions in the power sector by 2050 and scenario C builds on Scenario B by incorporating widespread electrification of the transport sector. The potential of switching heavy fuel oils with alternative liquid fuels is also explored in all scenarios. The modelling is conducted using a mixed-integer linear programming optimisation tool, the DECarbonisation Options Optimisation (DECO2). Results indicate that carbon emissions drop by almost 50 % if coal is replaced with biomass. Furthermore, a 100 % renewable energy grid is achievable with alternative liquid fuels generating around 1200 to 1500 GWh of electricity. In the case of net-zero emissions, 28 % of electricity can still be produced using fuel oils only if negative emission technologies (NETs) are deployed. Additionally, the electrification of transport is found to have significantly affected the electricity generation capacity. While the modelled system can meet a demand from electric vehicles of up to 400 GWh per year by the year 2050, the flexibility of choosing between the technologies reduces and expensive generation options need to be deployed. This electricity system modelling study aims at supporting policy decisions in national electricity grid planning in Mauritius.

1. Introduction

Evidence shows that continued emissions at the current pace will lead global temperatures to increase beyond 1.5 °C, causing severe and unprecedented damage to ecosystems and triggering multiple crises for the most vulnerable populations, including severe droughts and floods [1,2]. Mauritius is a part of SIDS, situated in the African continent, that faces disproportionate challenges and impacts from climate change [3]. Nonetheless, addressing this global challenge requires effort from all countries, although not of the same level [4]. In this regard, the Paris Agreement remains one of the strongest bonds uniting nations in keeping the global temperature increase below the threshold of 1.5 °C [5]. This overarching goal can only be accomplished when countries implement appropriate climate policies [6].

The power sector in Mauritius contributes over 50 % of the anthropogenic greenhouse gas emissions of the country [7]. The electricity grid infrastructure in the country consists of power plants owned by the CEB and IPPs. The CEB, a government-owned organisation, is responsible for the supply of electricity across the island to all consumers [8]. Mauritius is highly dependent on imported fossil fuels, mainly coal and fuel oils, to meet its electricity needs. In 2023, around 3270 GWh of electricity was produced collectively by IPPs and the CEB with less than 20 % from renewable energy sources, as illustrated in Fig. 1 [9]. Bagasse, a by-product of the sugarcane processing factories, is the major source of renewable energy. It serves as fuel during the crop season in thermal power plants operated by IPPs, which switch to coal outside the crop season. These coal-bagasse thermal power plants generate base-load power for the electricity grid. On the other hand, fuel oils are used in thermal power plants owned by the CEB and contributes to base-load in

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Abbreviations

AR	Afforestation and Reforestation
BECCS	Biomass Energy with Carbon Capture and Storage
CCS	Carbon Capture and Storage
CCUS	Carbon Capture, Utilisation and Storage
CDR	Carbon Dioxide Removal
CEB	Central Electricity Board
DECO2	DECarbonisation Options Optimisation
DME	Dimethyl Ether
EC-NETs	Energy Consuming Negative Emissions Technologies
EP-NETs	Energy Producing Negative Emissions Technologies
EU	European Union
EVs	Electric Vehicles
HFO	Heavy Fuel Oil
IEA	International Energy Agency
IPCC	Intergovernmental Panel on Climate Change
IPPs	Independent Power Producers
MARENA	Mauritius Renewable Energy Agency
MILP	Mixed-Integer linear Programming
MSW	Municipal Solid Waste
NDC	Nationally Determined Contribution
NETs	Negative Emissions Technologies
SIDS	Small Island Developing States
TIMES	The Integrated MARKAL-EFOM System
WASP	Wien Automatic System Planning

addition to serving semi-base-load [10].

In its updated NDCs, the government of Mauritius pledged to phase out coal and generate 60 % of electricity from renewable energy by 2030. For this purpose, a roadmap was developed: the 2030 Renewable Energy Roadmap which is henceforth referred to as the 2030 roadmap in this work. This document outlines the plans of the power sector for the period 2022 to 2030 but also raises significant concerns. Firstly, there is a substantial disparity between the current electricity generation system and the objectives outlined in the document. Moreover, the short timeframe of the roadmap limits the ability to set climate targets over longer terms, beyond 2030, for which substantial lead times may be required before actual implementation. For instance, while coal is planned to be phased out by 2030, a roadmap with a longer timeframe would provide insights into the future of fuel oils in the power sector which may be phased out as well after 2030. In this regard, biofuels such as DME have the potential to be used as a drop-in fuel in existing diesel engines [11]. At international level, investigation is being carried out to evaluate the possibility of replacing HFO with other alternatives; for example, Carvalho et al. [12] assessed the techno-economic viability of replacing HFO with green hydrogen and showed that a 100 % substitution is economically viable after 2030.

Another notable gap observed is that there is no indication that the 2030 roadmap was developed based on energy modelling. This was confirmed during a discussion with officers of the Ministry of Energy and Public Utilities of Mauritius. It is widely recognised that national energy planning decisions rely on energy planning optimisation models [13, 14]. To this effect, a wide range of opensource energy modelling tools exist [15] but selecting among the growing number of tools can be complex as their areas of application are not clearly defined [16]. For example, a commonly used modelling tool, WASP, only allows for cost-based optimisation of a power system over a period of 30 years [17], which impedes longer-terms modelling. More recently, the DECO2 has been developed as an open-access energy modelling software that optimises the evolution of electricity generation over a specified time horizon to achieve net-zero emissions using an MILP approach. One of the unique features of this model is that it allows users to incorporate NETs and fuel substitution strategies in existing power plants [18]. Another version of the framework has been developed to enable carbon trading in energy planning policies [19].

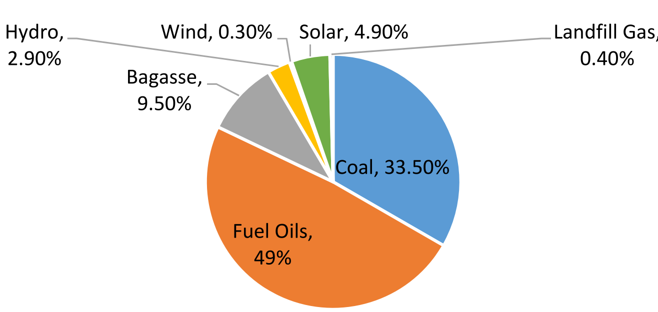

Fig. 1. Fuel mix for electricity production in Mauritius in 2023.

Internationally, several studies have developed long-term energy planning using optimisation modelling tools for the development of scenarios incorporating different climate targets. For instance, Vaillancourt et al. [20] explored the pathways for decarbonising Canada using the TIMES software but considered a 2 °C increase threshold rather than the more ambitious 1.5 °C. Another study by Burandt et al. [21] investigated the decarbonisation routes for China using an advanced version of the Global Energy System Model and demonstrated that the targets submitted in the country's NDC were insufficient to satisfy the aims of the Paris Agreement. Other optimisation models that account for cross-sectoral energy systems such as that developed by Sánchez Diéguez et al. [22] are particularly appealing for European energy systems due to the intertwining of various sectors and cross-country energy flows. However, such models require extensive data collection and detailed technology descriptions. Nonetheless, this high temporal resolution model underscores the importance of CCUS coupled with renewable energy in achieving negative emissions. The work by Quevedo et al. [23] is one of the few which conducted long-term energy planning in the context of SIDS. They used the open-source modelling software OSeMOSYS to explore the decarbonisation strategies of the energy sector of the Dominican Republic. Reunion Island, a neighbour island to Mauritius, analysed different scenarios to reach its target of reaching energy autonomy for electricity generation by 2030 using the TIMES optimisation model [24]. Gharehveran et al. [25] illustrated the potential of advanced optimisation techniques in enhancing the efficiency and reliability of modern power systems by developing an optimisation model in GAMS, structured as a mixed-integer quadratically constrained programming problem and solved using Gurobi solver. They demonstrated that solar PV farms integrated with energy storage systems and static synchronous compensators efficiently address power network problems such as voltage stability. One of the most recent developments in the arena of optimisation in the power sector involves the combination of Digital Twin model integrated with deep learning model and optimisation frameworks [26].

In Mauritius, existing literature has not efficiently demonstrated the application of modelling in the development of long-term power sector planning as agreed by MARENA [27] who is a key stakeholder in energy planning and policymaking in the country. For instance, Khoodaruth et al. [28] argued that a 100 % renewable energy system is feasible in Mauritius and identified different technologies to potentially deploy towards achieving carbon neutrality by 2050, but the lack of an evidence-based techno-economic analysis impedes the practicality of their work. Surroop [3] also proposed different pathways to decarbonise the energy system of Mauritius but failed to analyse the feasibility of the proposed strategies using mathematical modelling techniques.

Nonetheless, these studies are crucial in assessing decarbonisation strategies of the local power sector and should complement power generation modelling scenario formulations. On the other hand, Edoo and Ah King [29] used the EnergyPLAN software to investigate the possibility of achieving 35 % renewable share in power mix of Mauritius and focused on analysing the technical challenges of integrating a high level of variable renewable energy in the grid. While this work presents significant progress in understanding the complexities of integrating large amounts of variable power in the electricity grid, it is based on obsolete climate targets and considers a short time-frame of 2020–2025. Such short-term planning lacks practicality and the forward-looking perspective necessary for effective long-term policymaking [30]. Additionally, Timmons et al. [31] modelled the electricity system of Mauritius for a 100 % renewable energy grid in 2040. Although the study provides empirical ground for supporting a 100 % renewable energy source-based electricity system, the absence of gradual evolution of the energy mix impedes the decision-making process of policymakers in fixing achievable targets gradually over the span of the time-frame. Conversely, Hassen et al. [32] used the OSeMOSYS model to investigate potential fuel mix to replace coal as from 2030, yet excluded the strategies outlined in the 2030 roadmap. As such, the lack of effective studies on the long-term energy modelling of the power sector of Mauritius constitutes a gap in literature which is unfavourable in the development of long-term plans for the power sector.

Another significant observation is that none of the power planning studies mentioned above addressed net-zero emissions in Mauritius. The necessity for countries to transition to net-zero emissions by 2050 in view of keeping the global temperature rise to below the 1.5 $^{\circ}C$ threshold has been emphasised by the IEA [4]. The EU has already pledged carbon neutrality by 2050 [33,34]. Some SIDS are also envisioning net-zero emissions; Fiji aims for net-zero or even negative emissions by 2050 [35], while the Maldives is ambitiously targeting net-zero by 2030 [36]. In the context of net-zero emissions, Gaeta et al. [34] proposed roadmaps developed using the TIMES modelling software to achieve net-zero emissions in Italy by 2050 and argued that CCUS must be incorporated into biomass power plants (BECCS) to achieve negative emissions. The authors further stressed that any residual consumption of fossil fuels must be accompanied by CO_2 capture. This is consistent with the recommendation of the updated 'Net-Zero Roadmap' by the IEA [37] which mentioned that CDR technologies such as CCUS are required to achieve net-zero by 2050 across different sectors, including electricity generation. Also, both the IPCC [38] and the Energy Transitions Commission [39] shared the understanding that CDR mechanisms are necessary to prevent overshoot of the global temperature rise threshold. The effectiveness of removing carbon dioxide (CO_2) through NETs to decrease atmospheric concentrations of CO_2 has been demonstrated by Lemoine et al. [40]. Du et al. [41] also found that coal and biomass power plants in China post-2050 will need to be retrofitted with CCS to curtail emissions. In addition, Tan et al. [42] underscored the need for a variety of NETs through a balanced portfolio rather than individual NETs since climate targets cannot be achieved with a single option. It is important to note that different types of NETs are available, such as biochar, BECCS, algae culture and afforestation as demonstrated by Pires [43] who categorised them into direct and indirect air capture technologies. Nair et al. [18] on the other hand, classified NETs into EP-NETs and EC-NETs.

In addition, no studies were found that considered the power demand linked to electrification of the transport sector within long-term power sector planning for Mauritius although the use of EVs is being promoted through the implementation of several fiscal incentives [44,45]. Other island economies across the world are also encouraging the shift to EVs such as Trinidad and Tobago with similar fiscal incentives [46]. Locally, over $5,650,000 is anticipated to be invested in electric buses for public transportation during the period 2022–2028 [47]. MARENA [48] set out an ambition to phase out fossil fuel-based vehicles by 2040 which the prevailing 10-year roadmap for EVs does not consider. The document also does not technically assess the impacts of EVs on national electricity demand [49]. It is also worth noting that while electrification of land transport vehicles is important in limiting the rise in global temperatures [2,50], large-scale electrification of land transport will cause an increase in electricity generation as evidenced by Dixon et al. [51], which highlights the importance of its incorporation in long-term electricity planning.

Internationally, several studies were conducted on the electrification of transportation such as [23] which analysed the impact of penetration of electric mobility using the OSeMOSYS modelling tool and found that significant investment is required in the electrical power system. Yuan et al. [52] also conducted a detailed modelling of the road transportation sector and coupled the electrification of transportation with the power sector, using the EnergyPLAN simulation tool. Quevedo et al. [53] presented a set of open data to develop strategies in the energy transition of the land transport sector, supporting energy planning policies of island states.

Therefore, this study responds to the absence of a systematic application of mathematical modelling in developing Mauritius' long-term electricity generation strategies. It also aims to overcome the gaps in the existing modelling approaches including limited consideration of long-term climate ambitions such as net-zero emissions target. Additionally, it tackles the under-representation of EVs in long-term electricity planning. As such, long-term electricity scenarios for decarbonising the power sector of Mauritius are investigated using the DECO2 modelling framework with a cost minimisation objective. This modelling framework has been chosen primarily because it is an open-source software, allowing accessibility and adaptability for this study. Furthermore, it is relevant to developing countries and includes a wide range of technologies [18]. Additionally, unlike other traditional tools, DECO2 provides a robust optimisation capability, enabling comprehensive scenario-based analysis of energy systems, while incorporating a user-friendly Excel-based interface. This allows stakeholders from diverse sectors to engage with the modelling process without requiring advanced technical expertise. Furthermore, tools such as TIMES and OSeMOSYS focus on least-cost optimisation without explicitly aims for net-zero emissions [54]. On the other hand, DECO2 has been developed to allow exploring net-zero emission pathways for the electricity sector.

The long-term electricity model is developed for a timeframe of 2025–2050 which not only allows setting short and medium-term goals but also helps to envision future needs. Climate targets explored in the scenarios developed include the achievement of net-zero emissions by 2050 and the substitution of fuel oils with alternative liquid fuels in existing diesel engines, representing a novel contribution to existing literature. Electrification of transportation is considered by integrating the electricity demand forecasts from EVs in the model. While EV-related electricity demand is incorporated, the transport sector (e.g., modal shifts, EV and non-EV share, etc.) is not explicitly modelled because the modelling framework is limited to the electricity grid planning. The aim of this work is to support policymaking in decarbonising the power sector of Mauritius to inform policies and investments. The policy recommendations derived from evidence-based electricity modelling are applicable not only to Mauritius but also to other similar SIDS.

2. Methodology

The model for the power system of Mauritius was set up in the DECO2 framework which optimises the power generation expansion using MILP to satisfy electricity demands over the timeframe of 2025–2050, with a temporal resolution of one year, at minimum electricity system costs and specified emission targets. The toolchain of the framework is presented in Fig. 2 which consists of an integrated Excel-based user interface. The MILP model is built in Pyomo using Python programming language. Input data by the user is inserted in the Excel file and imported into Python using the Pandas library. This data feeds

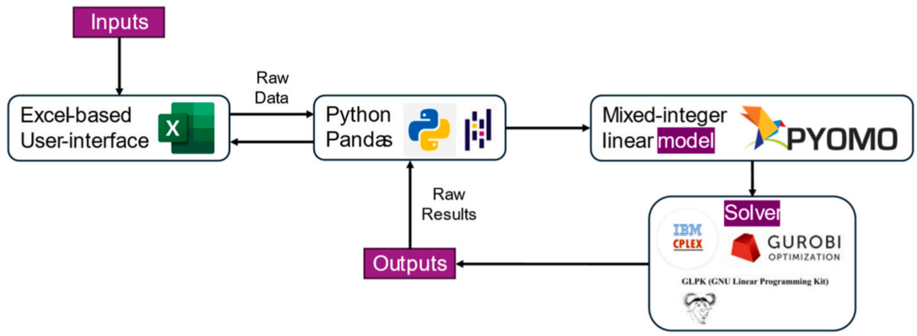

Fig. 2. The toolchain used in DECO2 model.

into the model formulated in Pyomo. The model is solved using the Gurobi solver. The raw results are then processed in Python and exported to the Excel user interface.

2.1. Modelling the electricity system of Mauritius

The power sector model was developed with the portfolio of existing power plants obtained from the latest annual report of the CEB [55] as well as the planned power plants outlined in the 2030 roadmap [10]. The portfolio of power plants in the model includes technologies for solar energy, wind energy, MSW, marine energy, HFO and solid-fuel power plants that can burn biomass and coal.

2.1.1. Solar energy

Mauritius being in the tropic of Capricorn, is favourable to harvest solar energy. The island receives from 5.5 to 8.5 h of bright sunlight daily [56]. Solar energy is already being exploited and an expansion of this technology is planned according to the 2030 roadmap. However, one of the caveats of large-scale deployment of solar power plants is the impact on land use and due to the land scarcity challenge in Mauritius [57], this study does not consider any additional solar energy technology deployment other than what has been provided in the 2030 roadmap.

2.1.2. Wind energy

Currently, the country produces only around 0.4 % of electricity from onshore wind turbines [58] and there is potential for expansion [59]. Moreover, while offshore wind energy is not being exploited yet, some studies have demonstrated its potential in Mauritius such as [60,61]. According to the latter, offshore wind turbines can yield above average energy potential. The 2030 roadmap projects the implementation of 50 MW of offshore wind turbines.

2.1.3. MSW

Several studies have demonstrated the huge potential of waste to energy from MSW [62–64]. At present, around 0.3 % of electricity is produced from MSW [65] and according to the 2030 roadmap, an additional 10 MW capacity from waste to energy will be implemented to generate 75 GWh of electricity as from 2029.

2.1.4. Marine energy

The 2030 roadmap envisions the development of up to 20 MW of marine renewable energy capacity by 2030. While marine renewable energy technology in Mauritius is currently non-existent, a few research efforts have explored its potential such as [66,67]. According to Ref. [61], wave energy in Mauritius has an average energy potential yield.

2.1.5. Hydropower

Hydropower is the oldest source of power in the country and today contributes around 3.6 % of electricity generation from the same. It is reported that the potential for hydropower has been fully exploited [68] and the 2030 roadmap also does not envision any additional capacity of hydropower.

2.1.6. Biomass

DECO2 also allows the substitution of coal with alternative solid fuels in existing power plants. Accordingly, the model was allowed to select alternative solid fuels in lieu of coal in the existing coal-bagasse power plants as from 2030 to align with the country's commitment to phase out coal as outlined in its updated NDC [10]. Here, it is assumed that biomass is the alternative solid fuels that will replace coal, hence, the coal-bagasse power plants will be converted to 100 % biomass plants as from 2030, operating on bagasse during the sugarcane crop season and on another locally available biomass outside the crop season or imported biomass. Timmons et al. [69,70] listed the different potential biomass sources that can be utilised. This approach is supported by Surroop [3] who stated that while an increased use of biomass is crucial in phasing out coal, firing locally available biomass such as cane trash, eucalyptus and forest waste in the existing power plants will avoid heavy capital investment. In fact, Reunion Island has also converted its coal power plants into 100 % biomass plants since 2023 and now operates on bagasse and wood pellets alternately [71–73]. Another advantage of this approach is that these thermal power plants can maintain electricity production for base-load demand.

2.1.7. Heavy fuel oils

Decarbonising the power sector also involves addressing the role of fuel oils which currently account for approximately 50 % of the electricity generation mix. Although no formal commitments have been made yet, the discussion with officers from the ministry of energy in Mauritius indicated that the option of using liquid biofuels in diesel engines is currently being explored. This was also suggested by Khoodaruth [28] who proposed to retrofit existing diesel engines using HFO so that locally produced bioethanol can be used instead of fuel oils. According to the authors, approximately 470 GWh of electricity can be produced from the total annual ethanol production from the local sugar cane industry. Besides, according to Elahee [74], a power plant was envisaged to be implemented that would operate on fuel oil at the beginning before shifting to biodiesel. Moreover, Timmons et al. [31] also considered bioethanol as a substitute for fuel oils in the diesel engines in their energy model.

Therefore, the model in the current study was allowed to switch fuel oils with alternative liquid fuels of lower carbon intensity as from 2035. However, due to the uncertainty regarding the availability and/or utilisation of alternative liquid fuels at present, this work investigates both cases; one when alternative liquid fuel is available and another when it is not available.

The detailed input data on the availability of each energy source across the years during the timeframe as well as the different costs of each technology are provided as Annex A. Regarding the costs, that of renewable energy are currently higher than fossil fuels except for biomass [61]. However, Timmons et al. [69] mentioned that the price of renewable energy is expected to decrease over time unlike that of fossil fuels. Amongst the renewable energy sources, marine energy and offshore wind energy have the highest costs [61]. Furthermore, the formulation of the model has been provided as Annex B and the open-source DECO2 software is available at its GitHub repository provided in the Digital Supplementary Information section.

2.2. Electricity demand

This study uses a temporal resolution of one year which allows for a precise timing of infrastructure investment decision for the relatively small, centralised electricity grid system of Mauritius. Due to the unavailability of forecasted electricity demand beyond 2030, an extrapolation was conducted using a growth rate of 1.5 % from 2030 to 2034 [69]. As from 2025, it is envisioned that energy efficiency programmes would largely influence electricity demand in the long term, curtailing the growth as seen in the case of the United Kingdom [75]. However, increasing economic activities, digitalisation and electrification [76,77] are expected to cause the demand for electricity to continue to increase and peak around 2050. Therefore, a declining growth rate every five years as from 2035 is adopted, i.e., 1.4 % growth in electricity demand from 2035 to 2040, 1.3 % from 2041 to 2045 and 1.2 % from 2046 to 2050. The demand profile for the period 2025–2050 is provided in Annex A.

2.3. Scenarios

Three scenarios are developed to evaluate the possible pathways for decarbonising the power sector.

2.3.1. Scenario A: low mitigation effort

Scenario A depicts a low mitigation effort scenario that considers only the existing target of phasing out coal by 2030 and the possibility of switching fuel oils with alternative liquid fuels as from 2035. The carbon budget in the model was based on the minimum feasible carbon budget which was determined by initially relaxing the carbon budget constraint and progressively tightening it until infeasibility was reached. The constraint was set at 2500 kt per year initially, representing emissions from the power sector in Mauritius [7]. The minimum achievable carbon budget obtained was 1200 kt CO_2 per year when an alternative to fuel oil was not available and 500 kt CO_2 per year when alternative fuels are made available.

2.3.2. Scenario B: net-zero system

In scenario B, net-zero emission is envisaged by 2050. The timeline for achieving this target is aligned with other countries including a few island states such as Malta [46]. Since the model gave infeasible solutions beyond the minimum achievable carbon budget found in the previous scenario, NETs are made available as from 2045 which allow the carbon budget to be constricted further as from 2045 to gradually reach 0 kt per year in 2050. In addition, like scenario A, a case is developed within this scenario to investigate the influence of the availability of alternative liquid fuels for replacing fuel oil as from 2035.

2.3.3. Scenario C: transport electrification

In this scenario, the net-zero target by 2050 is maintained and the uptake of electric passenger vehicles in the power sector is considered. Passenger transport consists of cars, dual purpose vehicles, vans and buses. As of July 2024, the country had a total of 2794 electric cars, 3 electric buses and 53 electric vans [78]. The average electricity consumption is taken as 0.1902 kWh per km for an EV car [79] and 1.57 kWh per km for an electric bus [80]. The average travel distance is 13, 500 km per year for cars and 38,000 km per year for buses [81]. Hence, by extrapolation, the number of electric passenger vehicles for the year 2025 is assumed and the electricity demand is calculated. As mentioned previously, MARENA [48] aims to phase out fossil fuel vehicles by 2040. In view of this ambition, this study explores the possibility of converting the entire fleet of passenger vehicles into EVs. Hence, the power demand for the complete electrification of passenger vehicles is inserted into the DECO2 framework. The equation for electrification of transportation has been included in the MILP model formulation for this scenario attached as Annex B. It is important to note that this scenario considers electricity demand resulting from EV adoption into long-term electricity demand projections. While this allows assessment of how transport electrification influences electricity generation planning, the study does not attempt to model the wider transport sector.

3. Results and discussion

This section presents the optimised results for all scenarios along with their detailed analysis.

3.1. Scenario A: low mitigation effort

Fig. 3 shows the temporal evolution of the fuel mix under the scenario A pathway without alternative liquid fuels. First, the validity of the optimisation results is evaluated. The cost of the power generation to meet demand in the first period (2025) is assessed. The modelling of scenario A resulted in a cost of $308 million to meet a demand of 3384 GWh for the year 2025. If the average price of electricity is $0.13 per KWh [8], the revenue derived from the sales of electricity for the year 2025 is $440 million. On the other hand, the actual revenue in 2022 was reported as $336 million, with 2698 GWh sold [8]. Since the difference is considered reasonable, the results obtained by the optimisation are considered valid for further analysis.

Additionally, the climate targets pledged in the NDC are satisfied as the optimised fuel mix resulted in around 67 % of electricity generated from renewable energy and coal being phased out by 2030. The fuel mix presents a high dependence on biomass as from 2030 as coal is replaced and represents more than 40 % of electricity generation. Similar results were obtained by Selosse et al. [24] for the neighbouring Reunion Island whose model also considered switching coal with locally produced biomass outside the sugarcane crop season. Also, the contribution of biomass for 2040 is similar to that obtained by Timmons et al. [69].

Another observation is that the fuel oils remain an important fuel source in the electricity mix and are expected to remain a key source for base- and semi-base-load electricity production. The optimised fuel mix contains a high share of fuel capable of delivering dispatchable power which is desirable since variable energy sources influence functional properties of power systems such as frequency and voltage response [82]. The inherent characteristic of variability or intermittency of renewable energy sources presents a major technical challenge to the local power system [83]. Although detailed power grid simulations are not within the scope of this study, the importance of grid capacity in accommodating higher shares of renewable energy is acknowledged. Nonetheless, in the present study, it is assumed that future investments will be made in the grid infrastructure to accommodate renewable energy integration beyond the current limits.

It is also observed that hydropower remains consistent throughout the entire period, being one of the cheapest sources of power in the country and serves the base-load [84]. Additionally, solar energy is progressively deployed over the years which aligns with the prediction that a rapid rise in solar energy technology manufacturing and investment will be observed [85]. Although battery energy storage is outside the scope of the analysis of this work, its importance is not denied. The expansion of battery storage is essential to better align the output of solar energy technologies with energy demand patterns and technical challenges faced by the renewable energy technologies [86]. The potential of integrating distributed solar power systems and mobile battery storage devices can also be investigated in future works [87].

Furthermore, some nuances have been noticed between the

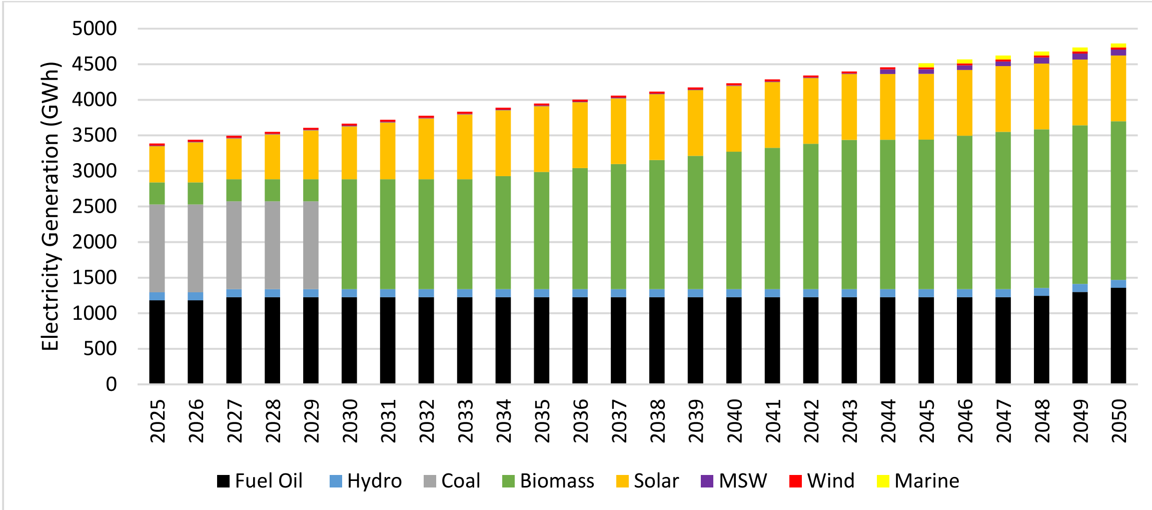

Fig. 3. Scenario A without alternative liquid fuels.

optimised fuel mix obtained and the plans elaborated in the 2030 roadmap. For instance, offshore wind energy is not deployed in this work in contrast to the 2030 roadmap. Also, implementation of marine energy and the expansion of the capacity of MSW energy occurs after 2030 while the roadmap proposed their use before this period. The delayed deployment of ocean-based energy technologies may be beneficial, as it allows for the implementation of more mature and cost-effective technology [88].

The fuel mix for the case when alternative liquid fuels are available to replace fuel oil in existing power plants as from 2035 is shown in Fig. 4. The model immediately switches to alternative liquid fuels which generated between 1200 and 1500 GWh of electricity per year during 2035–2050. In this case, the need for marine energy is eliminated. Notably, such a scenario represents a 100 % renewable energy system, aligning with Reunion Island's ambition for 2030 [89].

The carbon emissions for both cases in scenario A are presented in Fig. 5. It is observed that phasing out coal reduces carbon emissions by almost 50 % which is reasonable since Reyseliani et al. [90] claimed that phasing out coal would reduce the need of CCS by 71 %. Moreover, the IPCC's "Synthesis Report" [91] stated that CO_2 emissions must be halved by 2030 to limit global warming to 1.5 °C and this work demonstrated that Mauritius could achieve this climate target by phasing out coal alone. Phasing out of coal reduces emissions from approximately 2500 kt per year to around 1200 kt per year. However, emissions remain nearly constant throughout the remaining timeframe. The use of alternative liquid fuels instead of fuel oils could further lower emissions to about 500 kt per year. A notable deduction is that, despite a 100 % renewable energy-based electricity grid, net-zero emissions are not achieved as NETs are not available in this scenario.

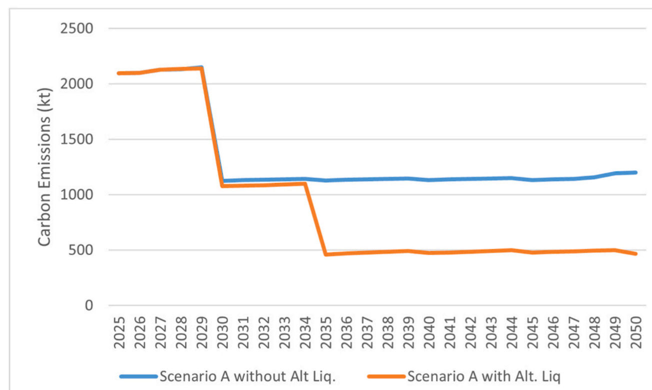

Fig. 5. Carbon emissions for scenario A.

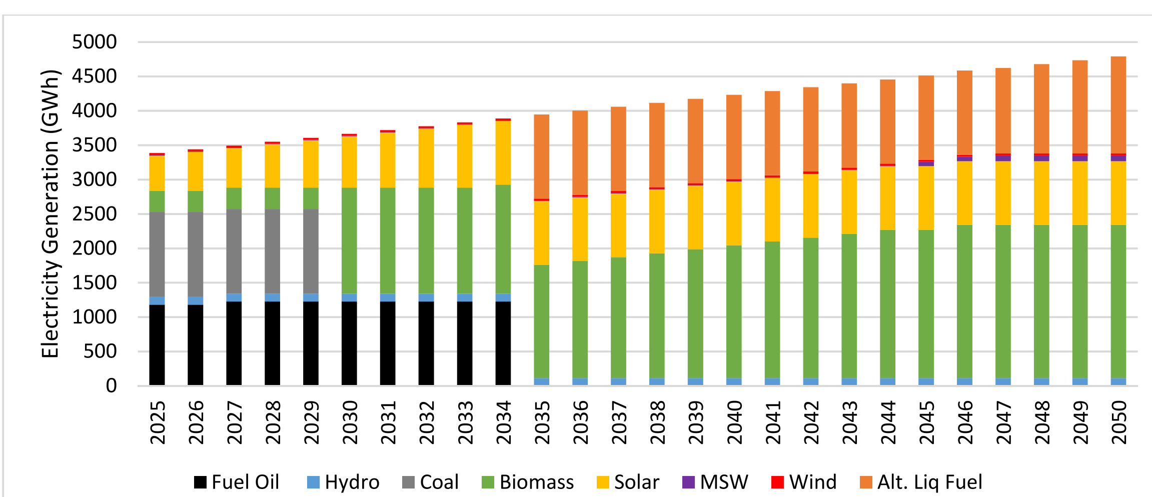

Fig. 4. Fuel mix for scenario A with the availability of alternative liquid fuels.

3.2. Scenario B: net-zero 2050

The fuel mix for scenario B, which envisaged achieving net-zero emissions by 2050, is shown in Fig. 6. Like the previous scenario, offshore wind energy is not utilised at any time during the timeframe while marine energy is deployed only as from 2045 with a relatively lower cost as the technology matures [18]. This is one of the main differences noted between the 2030 roadmap and this work, which further strengthens the importance of evidence-based optimisation modelling with a least-cost objective in long-term electricity planning.

Another significant analysis is that phasing out coal and the use of renewable energy technologies are essential steps but insufficient to achieve carbon neutrality by 2050. To achieve zero emissions, EP-NETs and EC-NETs were needed. This outcome is aligned with the results obtained by Quevedo et al. [92] for the case of the Dominican Republic.

The deployment of the NETs presented in Fig. 7 shows a progressive increase in capacity corresponding with the increasing demand for electricity and the decreasing carbon budget allocated in the model. A viable form of NET here would be biomass power plants fitted with CCS/CCUS, i.e., BECCS which act as net carbon sink [43]. Both Sánchez Diéguez et al. [93] and Cossutta et al. [94] supported such an approach for achieving ambitious climate policy like net-zero emissions. Also, the optimisation results shown in Fig. 7, suggest that the model chose to match the electricity demand of the EC-NETs with the electricity produced by the EP-NETs.

It is worth noting that, although net-zero is achieved in this case, around 28 % of electricity is still being produced from fuel oils in 2050. According to Fam and Fam [95], a net-zero electricity system can consist of a combination of renewable energy sources and fossil fuels along with a CDR technology. The IEA [96] also states that there are various pathways to limit the global temperature rise. Therefore, the result of this scenario may be reasonable in the absence of alternatives to the fuel oils-powered generators currently being used to meet the base- and semi-base-load demand of the grid and are thus, a critical element in ensuring reliability of power supply.

On the other hand, if alternative liquid fuels are available and utilised in existing diesel engines, fuel oils can progressively be reduced until completely phased out by 2045 as illustrated in the optimised fuel mix shown in Fig. 8. In this case, energy from MSW is not used at any time during the entire period while marine energy is started as from 2045. This indicates that the model now has more flexibility in choosing between the lowest cost fuels to meet the demand while satisfying the carbon budget allocated. Moreover, the capacity requirement for NETs is

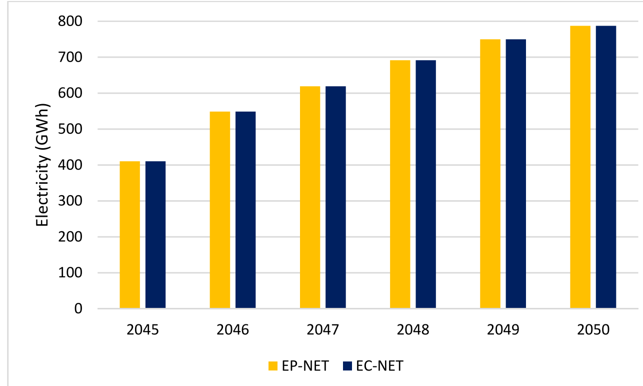

Fig. 7. NETs deployment for scenario B without alternative liquid fuels.

reduced as suggested by Fig. 9 since the electricity system uses fuels with lower carbon intensity.

Carbon emissions for scenario B are presented in Fig. 10 and clearly NETs from 2045 allows emissions to decline up to zero in 2050. Hence, it can be suggested that NETs offer flexibility to the electricity system to satisfy stringent carbon emission targets [40]. Besides, alternative liquid fuels also contribute significantly in bringing down emissions to less than 500 kt/y as from 2035.

This scenario confirms that NETs can play a crucial role in reducing emissions and achieving net-zero which is consistent with the published findings [97,98]. Barrett et al. [98] stated that CDR technologies are required for reducing emissions, although, the emphasis of their study was to support the need for reducing electricity demand which would reduce the amount of deployment required for CDR and decarbonisation.

3.3. Scenario C: net-zero transport

Scenario C explores the impact of electric mobility on the electricity mix of the country. As mentioned before, this study investigated the possibility of electrifying the entire fleet of passenger vehicles by 2040. However, this led to infeasibility. Consequently, the electricity demand from EVs was progressively reduced until a feasible solution was finally obtained which was corresponded to around 40 % of the passenger vehicle fleet. The initial infeasibility of the model was primarily due to

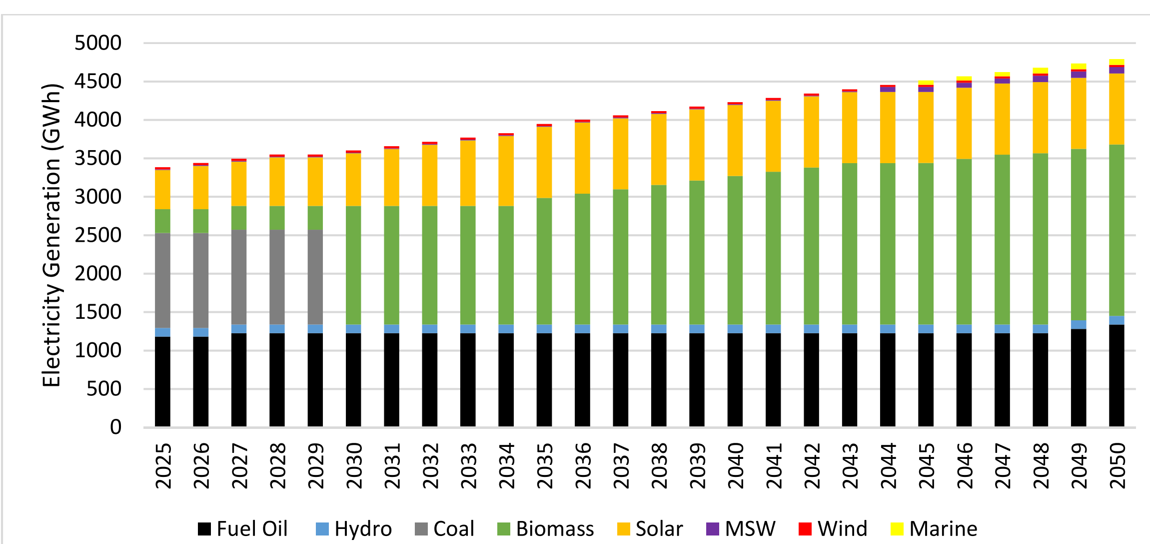

Fig. 6. Fuel Mix for Scenario B without alternative liquid fuels.

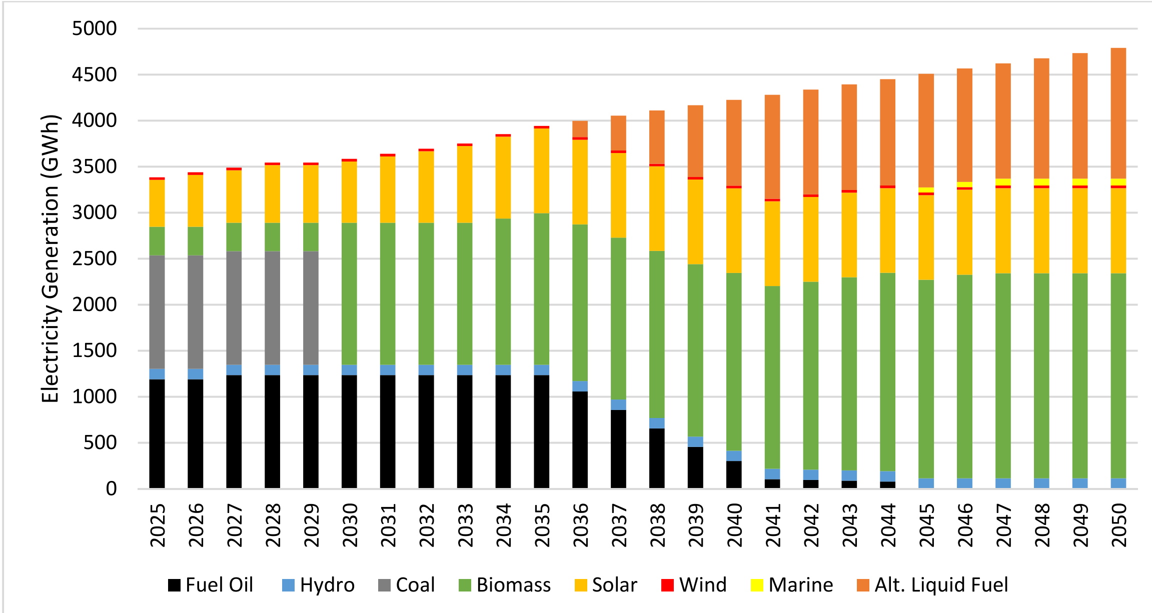

Fig. 8. Fuel mix for scenario B with alternative liquid fuels.

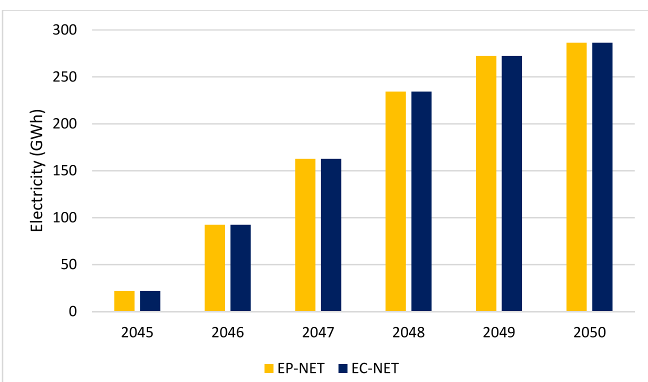

Fig. 9. NETs deployment under scenario B with alternative liquid fuels.

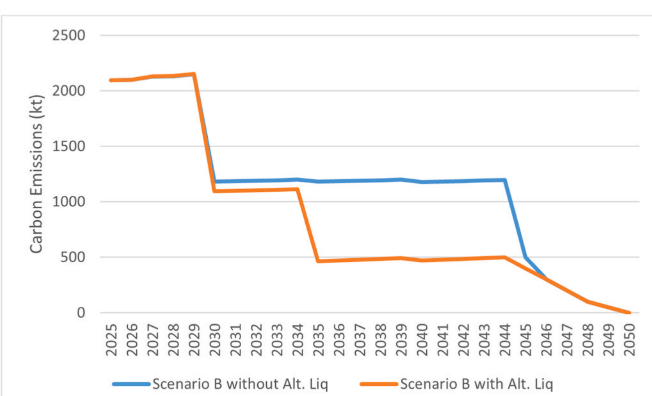

Fig. 10. Emissions for scenario B.

insufficient resource flexibility. More precisely, the electricity system, as configured in the model, exhibited a limited ability to adjust the supply in response to the demand and feasible solutions were obtained only after throttling. Godoy et al. [50] also obtained similar results and concluded that a high rate of electrification of land transport vehicles is only possible alongside a reduction in final energy demand. Furthermore, the multi-objective optimisation model proposed by Ref. [99] that incorporates multi-energy supply and flexible load is an interesting avenue for demand side electricity management since flexible load enables decoupling power demand from energy consumption. Their work showed that incorporating flexible demand significantly enhances system performance by reducing integrated costs, increasing the utilisation of renewable energy, lowering reliance on peak thermal power generation and decreasing the need for battery energy storage systems. Hu et al. [100] also presented similar results based on a two-layer optimisation model.

Therefore, using a more subtle transition of passenger vehicles to EVs, the electricity demand for EVs was assumed to progressively increase by 5 GWh per year from 2025 to 2044 and by 25 GWh per year from 2045 to 2050. Electricity demand from EVs by 2050 reaches 400 GWh per year. It is worth noting that the growth rates of the electricity demand used is aligned with the IEA's Net-Zero Emission by 2050 Scenario which stated that the average annual rate of growth of EVs is 27 % until 2035 [96].

The optimised fuel mix for scenario C is shown in Fig. 11 which suggests that the increased demand from the transport sector has reduced the flexibility of choosing between the supply options and even the most expensive technologies need deployment in order to satisfy the demand. This is in agreement with the finding of the IPCC which stated that reducing energy demand could provide greater flexibility in designing energy systems [91].

This scenario demonstrates that electrification of even a segment of the land transport sector has a considerable impact on the electricity demand of the country. Hence, ambitions for decarbonising the transport sector using EVs cannot be excluded in the long-term term electricity planning of the country and the use of a least-cost optimisation tool helps policymakers in making informed investment decisions. It should be noted that while the electrification of passenger vehicles caused an increase in electricity demand while emissions remain unchanged relative to scenario B since the climate targets were maintained.

3.4. Economic implications

The DECO2 framework provided the optimised fuel mix at minimised cost as objective function. The optimised costs of the electricity

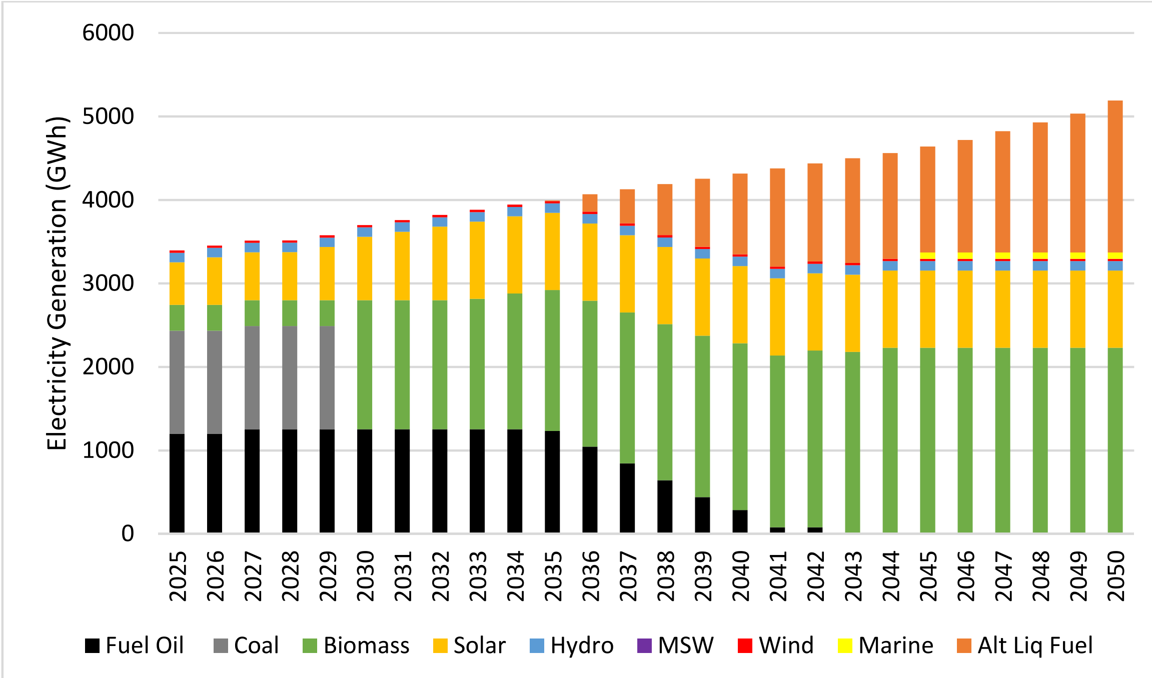

Fig. 11. Fuel mix for scenario C.

system for the different scenarios are illustrated in Fig. 12. Overall, there is a gradual increase in system costs for all scenarios which is due to the model deploying more generation capacity to satisfy the yearly increase in electricity demand. Notably, there is significant divergence in system costs between the different scenarios as from 2045 onward. Focusing on this timeframe, it is observed that while for both cases of Scenario A, there was a modest increase in costs relative to the previous years, however, the system costs for the other scenarios substantially escalated. A likely explanation for this outcome is the deployment of NETs as from 2045 in ambition to achieve net-zero emission targets by 2050, influencing the cost of the electricity system. Furthermore, the results of the modelling show that the replacement of fuel oils with alternative liquid fuels has a major cost implication since Scenario B without alternative liquid fuels recorded higher costs than Scenario B with alternative liquid fuels as well as Scenario C. This observed trend can be linked to higher NETs capacity requirement when alternative liquid fuels are not utilised, since the carbon intensity of the fuel mix is higher.

Additionally, it is interesting to note that the total cost for the period 2025–2030 in this work is around $1800 million while the 2030 roadmap estimated an investment of $1345 million for the period 2022–2030. The difference is considered reasonable since this study not only accounted the investment costs for new power plants but also considered the cost of fuel used in existing power plants. Also, the RE Roadmap 2030 may not have factored in the costs of coal and the alternative biomass as these fuels are used by power plants operated by IPPs.

3.5. Sensitivity analysis

Sensitivity analysis allows assessing the robustness and reliability of the results [101]. In addition to the different scenarios analysed, a sensitivity analysis is performed on electricity demand and carbon budget, which we determine as being the most sensitive and uncertain parameters for overall system performance. For the first case, the annual growth in electricity demand was increased by 0.1 % and then by 0.2 %. As expected, electricity generation is increased proportionately from the

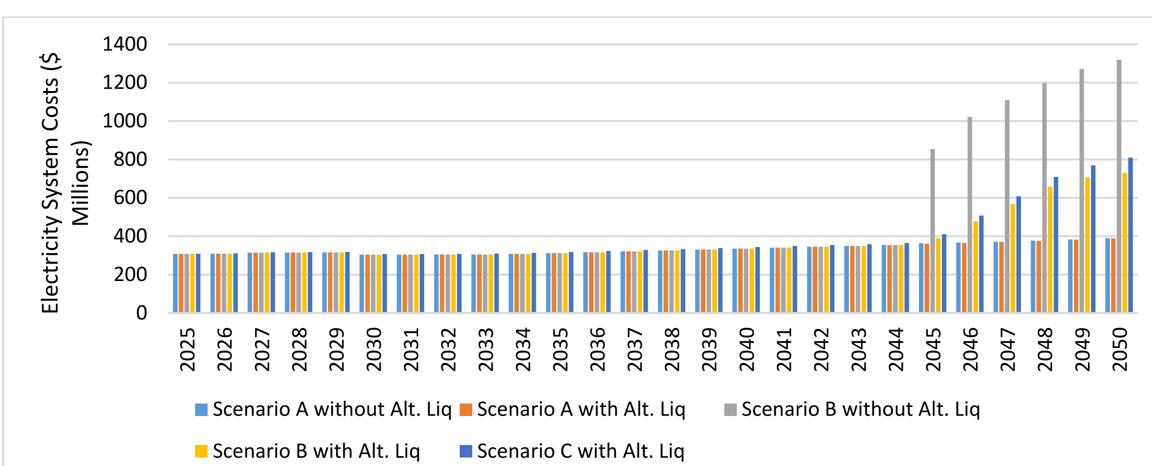

Fig. 12. Costs of the electricity system.

different sources to satisfy the demand while the overall system costs increase logically. Furthermore, when the annual growth in electricity demand was set to be 4.5 % as of 2025, reflecting the trend from the past few years, and gradually decreasing by 1 % every five years, the model gave infeasible solutions because of the insufficient supply capacity to meet this demand. However, if the demand stabilises from 2035 until 2050, feasibility is obtained only if the carbon budget remains high at 2500 kt/y or if the NETs are made available. Moreover, tightening the carbon budget by 10 % gives an infeasible result since the electricity system is not able to meet the demand with the available resources. Subsequently, when alternatives liquid fuels are made available from the first year, the model returns feasible results since it can now use available technologies to lower its emissions while meeting the demand. These results show that both electricity demand and carbon budget are significant determinants of feasibility. Also, the model reacts logically and predictably to variations such as infeasibility when a high demand has to be met along with a tight carbon budget, proving its robustness and utility in understanding the system behaviour.

4. Policy recommendations

Climate policies can directly influence temperature pathways, resulting in different levels of climate risks [1]. Some policy recommendations are proposed here based on the results from all the scenarios.

4.1. Energy efficiency and energy demand reduction

The results of the optimisation of the three scenarios demonstrated that energy efficiency is key to reduce demand and meet carbon constraints. For example, the results of scenario C showed that increasing electricity demand from the transport sector may lead to carbon budgets not being met. As such, policy priorities should be centred on reducing electricity consumption. Barrett et al. [98] demonstrated that energy demand reduction is crucial in reducing the costs of emission reductions and also recommended the creation of an ‘Energy Demand Reduction Delivery Plan’ which should contain energy demand reduction targets. A similar strategy is recommended for the electricity sector of Mauritius. Also, a wide range of studies underscored the importance of energy efficiency coupled with a shift in the energy consumption pattern in achieving net-zero [34,50,98,102]. A major contribution was made to literature by Gharehveran et al. [103] who formulated a mixed-integer non-linear programming problem and employed a multi-objective optimisation algorithm, minimising both operational costs and carbon emissions. Their work primarily emphasised the strategic use of demand response programs in enhancing micro-grid performance and presented a novel incentive-based payment strategy package as pricing offer to a diverse range of consumers. The authors demonstrated that their strategy led to a 12.5 % cost reduction and 14.3 % decrease in emissions with respect to baseline scenarios.

4.2. Alternative liquid fuels

One of the novelties of this study is considering the replacement of fuel oils with alternative liquid fuels which gives the model greater flexibility in selecting the least cost fuel mix while satisfying narrow carbon budget constraints. Furthermore, the replacement of fuel oils with alternative liquid fuels not only reduces emissions by more than 50 %, but it also lowers the costs of the system in the case of net-zero emissions. This study demonstrates that the potential of switching fuel oils with other alternative liquid fuels of low carbon intensities, however, it is acknowledged that infrastructure constraints concerning local production or imports of alternative liquid fuels such as biofuels, have not been considered in this study. It is recommended that this should be investigated in depth in future research. Nonetheless, achieving climate targets entails the adoption of emerging technologies since innovative technologies currently being developed will provide the possibilities to more emissions reductions [104]. Policymakers should remain open to innovative technologies as net-zero will principally be geared by innovations and the gradual decommissioning of old technologies [105,106]. The importance of long-term energy planning is strengthened here since it allows setting goals ahead of time. Therefore, it is recommended that the next priority after phasing out coal should be eliminating fuel oils and achieving net-zero emissions.

4.3. Just transition

It is also noted that the phasing out of coal and fuel oils will affect employment in the energy sector [107]. In China, Zhang et al. [108] found that for every 1 % growth in clean energy production there is a 0.013 % increase in total employment and that job losses are primarily focused in coal extraction. Yuan et al. [109] found that wind and solar power have the potential to create more job opportunities to offset job losses caused by the phasing out of coal. Additionally, Osorio-Aravena et al. [110] showed that a 100 % renewable-based energy system would engender significant socioeconomic benefit in Chile. Meanwhile, an equity-based priority system has been proposed for the retirement of coal-fired power plants in Australia which integrates several factors including employment and electricity price [111]. However, in the present study, coal is phased out by retrofitting the existing power plants to operate entirely on biomass rather than retiring them. While recognising that the decarbonisation may have an impact on employment, this analysis is outside the scope of the study. However, it is highly suggested that future research investigates the impact of the decarbonisation on local employment to ensure a smooth equitable energy transition.

4.4. Negative emission technologies

There is a need to analyse the implications of the adoption of NETs by Mauritius in the near future. It is widely recognised that SIDS suffer from disproportionate disadvantages in the context of climate change since they bear the risks of facing the most severe impacts due to their particularities while having contributed the least to this global phenomenon [112–114]. On top of that, requiring SIDS to further stretch their ongoing mitigation and adaption efforts to implement NET may present a significant moral concern, given the high investment costs necessary and other resource requirements. Even if NET projects are funded by third parties from developed countries, their local implementation may still exacerbate the existing scarcity of land resource pressure in SIDS as many studies pointed out that technologies like BECCS and AR require significant land [115,116]. This recent study [117] highlights the challenge of land use in Mauritius, which needs to be considered.

Consequently, the decisive question is whether the net-zero emissions pathway should be adopted by Mauritius, considering that NETs are a necessary part of this journey, or whether emphasis should be placed on other climate targets that are less stringent for the country. Besides, in its latest updated Net-Zero Roadmap report, the IEA [85] emphasised that the pathways elaborated in its report allows emerging and developing countries more time to reach net-zero in comparison to advanced economies and that not all countries are required to achieve net-zero by 2050. For policymakers to effectively address the decisive question, it is suggested that deeper insights about the implications of each scenario are gained such as resource requirement and investment potentials. Recently, Weerdt [118] assessed empirically the moral hazards associated with delayed investment in emission reduction technologies including NETs and concluded that it is crucial for policymakers to consider the dynamics involved in policymaking and technology adoption.

4.5. Decarbonisation of the transport sector

An important analysis that emerged from scenario C is that the current electricity system infrastructure will not be able to sustain a complete electrification of the passenger vehicles in Mauritius. In addition, while currently there are a few public charging facilities for EVs, the public charging infrastructure must be analysed for increase in the EV fleet since several studies have highlighted that electric mobility faces lack of public charging facilities [119,120]. This will also come at an increase in costs, not currently analysed in our model. It is further recommended that the electrification of transport sector is carried out through smart charge strategies [121] to maximise the amount of managed charging compared to unmanaged charging. For example, electricity tariffs can be designed to align charging decisions that supports minimising the power system costs [122]. It has been highlighted in the literature that the effective energy management may prevent EVs' charging during the peak periods to avoid severe voltage reduction [123].

However, the complete decarbonisation of the transport sector as set out by MARENA [124] would necessitate the implementation of other strategies, based on the results of this study. According to Burandt et al. [125] electrification along with a large-scale adoption of biofuels are viable alternatives for replacing conventional fossil-fuelled based vehicles when considering strict carbon budgets. In fact, the use of biofuels for land transport has remained a debated topic in the local policy-making and research community for more than a decade [126]. Hydrogen is also expected to play a significant role as a synthetic fuel helping to decarbonise the transport sector as from mid-2030 [34,85,127–130].

4.6. Cross-sectoral cooperation

Achieving the overall goal of decarbonisation will require cross-sector coordinated efforts for creating policies and strategies in transportation, energy and environment. For instance, fiscal initiatives such as tax exemptions for EVs and hybrid vehicles may be required for electrification of the transport sector [130]. Several studies have found that decarbonisation of the power sector has an impact on other sectors and subsequently, requires cross-sectoral strategies. Reyseliani et al. [90] found that cross-sectoral coordination is important to provide to support the decarbonisation of the power sector. Raillard-Cazanove et al. [131] also demonstrated that decarbonisation of the electricity system is linked to the reduction of emission in the industry.

5. Conclusion

This study presented pathways for decarbonising the power sector of Mauritius using mathematical modelling and optimisation. Different scenarios representing the decarbonising pathways, were formulated for the 2025–2050 timeframe. Scenario A portrayed an electricity system consisting of only the 2030 climate targets pledged in the NDC by the government of Mauritius, i.e., phasing out coal and generating at least 60 % of electricity from renewable energy sources by 2030. It was assumed in this study that coal will be replaced with local biomass such that the existing coal-bagasse thermal power plants would operate on bagasse during the sugarcane crop season and on another local biomass outside the crop season. Consequently, the optimised fuel mix showed heavy reliance on biomass and gradual expansion of the other renewable energy technologies over the timeframe. Also, it was revealed that carbon emissions will be reduced by more than 50 % from its current level by 2030 when coal will be phased out, that is, from 2500 kt per year to around 1200 kt per year. It was also found that alternative liquid fuels can further reduce the carbon budget to around 500 kt per year by 2050 and will eliminate the need for marine energy.

Scenario B explored a decarbonisation pathway with a target of achieving carbon neutrality by 2050, in addition to the 2030 targets pledged in the NDC. This scenario evidenced that phasing out coal and the deployment of renewable energy technologies are essential but will not be sufficient to achieve carbon neutrality alone which requires deployment of NETs. The optimised fuel mix again showed heavy reliance on biomass, and it was suggested that BECCS would be a viable NET that can be implemented. Due to the integration of NETs, the costs of the electricity system were higher than the previous scenario. Interestingly, it was found that replacing fuel oils with alternative liquid fuels would reduce the costs considerably since the required capacity of NETs was reduced. This demonstrates that alternative liquid fuels will be pivotal in the local energy landscape and it is recommended that the phasing out of fuel oils constitutes the next priority along with net-zero emissions after the 2030 climate targets submitted in the NDC.

The impact of the uptake of passenger-EVs on the electricity system was analysed in scenario C in addition to the 2030 targets and the net-zero target by 2050. The results showed that electrification led to increased demand for electricity which reduced the flexibility of the model to choose between the least cost fuels. Also, only 40 % passenger car fleet electrification would be feasible with the electricity generation portfolio outlined in the 2030 roadmap. The scenario demonstrates that electrification of the transport sector has a considerable impact on the electricity demand of the country and as such, decarbonisation of the transport sector through massive deployment of EVs should be integrated in the long-term term energy planning of the country.

Consequently, this study not only addresses significant gaps found in literature for electricity modelling of the Mauritian electricity sector but also explores pioneering approaches to decarbonise the electricity sector. This work is aimed to support to policymakers in developing an evidence-based long-term electricity plan for Mauritius. The report can also serve as a framework for studies in other SIDS.

Future studies could focus on enhancing the temporal resolution to capture the operational constraints of high-renewable power systems. The DECO2 modelling for Mauritius power sector could also be expanded to incorporate carbon pricing and life cycle assessment to evaluate land use and potential carbon trading policies [132]. Additionally, an extension to the model could be envisaged to include socio-economic impacts such as job creation and energy affordability which ensures alignment with just energy transition principles. While the study presented several innovative elements for the power system such as switching HFO with alternative liquid fuels, it also assumes that the infrastructure is adaptable. Recognising the technical, financial and regulatory challenges of these transformations, it is recommended to incorporate infrastructure constraints and sensitivity analysis (on carbon prices, infrastructure costs, land use, etc.) to assess the resilience of Mauritian power sector. It is also recommended to explore the cybersecurity-related issues and the reliability of power networks [133] because the future electricity grids may have a portfolio of power generation, energy storage and emission reduction technologies managed by computational tools.

Digital supplementary information

The DECO2 software is available at its GitHub repository (https://github.com/mchlshort/DECO2) and website (https://institute-for-sustainability.turtl.co/story/deco2/page/1).

Credit author statement

Nazeefah Edoo: Conceptualisation, Methodology, Data acquisition, Investigation, Software, Funding acquisition, Writing-original draft & review & editing. **Gul Hameed:** Supervision, Software, Methodology, Data acquisition, Writing - review & revised. **Michael Short:** Conceptualisation, Methodology, Supervision, Writing - review & revised.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nazeefah Edoo reports financial support was provided by Chevening Awards in partnership with Transforming Energy Access. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.esr.2025.101930.

Data availability

We have uploaded all the data used as supplementary information in the Annexes

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